receipts and payments. Both users and customers are instantly notified of their transactions digitally over their smartphones. The advantage lies in the fact that enterprises can download any transaction receipt copy from the cloud at any time for no cost. Consequently, the lesser the involvement of physical paperwork, the safer and the easier to manage the transactions. Carrying and maintaining a large amount of cash raises security concerns. Additionally, keeping a track of all transactions on paper becomes very difficult. Digital management and transaction omits all such troubles, keeping a safe, organised and well-recorded track of all your money. EFY

—Paromik Chakraborty, technical journalist, EFY

PIR Sensors To Reduce Electricity Wastage

› PIR sensors automatically turn off electric appliances when not in use

It's eight o'clock at night. Everyone has left for the day. Still, you see all the lights and electric appliances running in your office. Putting up signs like "last person to leave turns off the lights" is read and followed by only a handful. Keeping a staff just for turning off the appliances when not in use is also not a cost-effective solution. So how to get around it? Energy-efficient lighting like light-emitting diodes (LEDs), intelligent technologies like smart control and automation, and more of the likes have been the talk of the town for a while. There is one more very competent candidate in this mission of energy savings that goes by the name of passive infrared (PIR) motion sensors.

Be it a small office or a large industrial plant, a substantial portion of the whole floor remains vacant throughout the day while the appliances keep running, wasting a great deal of energy. PIR sensors detect movements in an area based on the infrared heat signatures emitted by warm bodies. Depending on the infrared reading, appliances are switched on or off.

Slash your monthly electricity bill. Use of PIR motion sensors in your office space can cut your electricity bill by up to 30 per cent, depending on factors like the area covered, the frequency of electricity usage in the covered area, load devices that the sensor is connected to and so on. PIR sensors themselves draw less than one watt of power per hour in idle situation, so their own electricity consumption does not impact the bill at all. PIR sensor manufacturers and dealers in the Indian market claim a cost saving of up to 30 per cent if PIR sensors are used properly for electric appliances in offices. For instance, lights or fans in meeting rooms, conference halls, washrooms or other areas keep running even when the rooms are unoccupied. These are areas where major savings can be achieved.

According to Delhi-based Highly Components Electronics, if a 40W tubelight used in a corridor runs for 12 hours a day, its operating time can be reduced to just three hours. Considering the average electricity cost of ₹ 4.5 per unit consumed, monthly cost of running one such tubelight can be cut down from ₹ 64 to ₹ 16, i.e., a monthly saving of ₹ 48 per tubelight.

For instance, Hyderabad-based Wells Fargo India Solutions has

An outdoor bulb controlled by a PIR sesnsor (Image courtesy: Wikimedia Commons)

been able to cut energy usage in its unoccupied zones by 10 per cent and overall energy consumption by 30 per cent using Wipro Lighting's sensors. The solution included occupancy detector installation and personalised lighting control among others.

A green premise and quick ROI. Cost cutting using PIR sensors implies a quick return on investment. Swati Deshpande, CEO of Shriyuj Electro Controls, says, "On an average, it will take up to one year to get a return on investment in PIR sensors for lighting. It however, depends on a few factors like the area of your office space, the sitting arrangement, the number of sensors required and so on." The technology also drives enterprises towards an environment-friendly foundation. Lower electricity wastage results in lower carbon gas emission, reducing a company's carbon footprint significantly.

Swati shares an elaborate case study in this aspect. Belgium-based Bekaert, a steel-wire transformation and coatings manufacturer, was